

Stoquify Enterprise HRIS Current State And Full System Proposal

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Repository: `E:\ohada_saas\Focused_projects\stoquify`

Prepared from: saved execution prompt, current codebase inspection, HRIS/payroll roadmap documents, UI documentation, and specialist review lenses covering HR operations, UX, compliance, security, data architecture, product scope, and release readiness.

1. Executive Verdict

Stoquify already contains a serious payroll-grade HR source layer. It has employee master records, effective-dated contracts, compensation controls, payment-destination change evidence, attendance snapshots, payslip self-service, payroll readiness checks, certified input proof, payment/declaration proof, and close-assurance hooks. These are valuable foundations.

It is not yet a full enterprise HRIS.

The current implementation is best described as a payroll-controlled HR source foundation embedded inside the payroll domain. It is suitable for controlled pilot or limited release of implemented, evidence-gated payroll workflows where the scope is explicit. It should not be marketed or operated as unrestricted production HRIS/payroll until the missing HRIS boundary, HR lifecycle workflows, retention controls, statutory certification, live integration evidence, manager/employee self-service, browser/accessibility evidence, and release attestation controls are complete.

The strongest strategic move is not to rewrite the payroll kernel. The right move is to establish a dedicated HRIS/People boundary that owns people truth, then make payroll consume certified HRIS snapshots. This aligns with the existing roadmap language: HRIS owns people truth, payroll consumes certified HRIS snapshots, accounting records money truth, and assurance proves the chain.

2. Scope And Method

This report evaluates:

- What HRIS functionality exists now.
- What is missing for a professional, modern, enterprise-grade, secure HRIS.
- Which current payroll tables/services should be reused, wrapped, or replaced.
- How the future HRIS should fit Stoquify's dark command-center dashboard design system.
- Which security, privacy, audit, compliance, and release controls are required before broader production use.
- A staged implementation path that avoids unnecessary rewrites.

Primary evidence inspected included:

- `prisma/schema.prisma`
- `services/payroll/*`
- `actions/payroll/*`
- `components/payroll/*`

- `config/permissions.ts`
- `lib/security/rbac-permissions.ts`
- `services/security/redaction-policy.service.ts`
- `services/accounting/*close*` and data-trust evidence surfaces
- `docs/HR-Payroll/*`
- `what-next/payroll/*`
- `docs/UI/UX/*`
- `docs/product/user-experience/ui-registry.md`
- `components/dashboard/primitives/command-center-primitives.tsx`

No production code was changed for this report.

Specialist review inputs were incorporated from HR operations, UX architecture, compliance/audit, security architecture, database architecture, and product strategy perspectives. The report uses completed specialist findings plus direct repository evidence.

3. Confirmed Current HRIS-Like Capabilities

3.1 Employee Master Foundation

The `PayrollEmployee` model is the current employee master record. It includes organization scope, optional user mapping, employee number, display/legal names, status, hire and termination dates, country, location, department, job title, cost center, masked/hash tax identifiers, masked/hash social identifiers, masked/hash payment destination fields, and relations to contracts, compensation assignments, payment-destination requests, attendance snapshots, run lines, payslips, allocations, and balance cases.

Evidence:

- `prisma/schema.prisma:1644` defines `PayrollEmployee`.
- `services/payroll/employee.service.ts` enforces tenant-scoped employee operations, user/location validation, duplicate mapping checks, evidence hashes, audit behavior, redaction policy, and readiness blockers.
- `actions/payroll/payroll-employee.actions.ts` routes employee operations through payroll permissions and fresh-auth policies for sensitive operations.

Assessment:

This is a credible starting point for an HRIS employee identity layer, but it is still named and owned by payroll. A full HRIS should make employee identity a People/HRIS-owned boundary, even if it initially wraps this table.

Data caution:

The current employee record also conflates identity, employment status, organization placement, identifiers, and current payment destination. The future HRIS should avoid copying this into a second employee master. During transition, use the current employee row as compatibility storage, make HRIS services the only writer, and move payroll-affecting facts into effective-dated HRIS-owned version tables.

3.2 Contract Lifecycle

The `PayrollContract` model supports employee contracts with organization scope, effective dates, status, type, salary, currency, working hours, classification, convention, signed document hash, and business-event linkage.

Evidence:

- `prisma/schema.prisma:1701` defines `PayrollContract`.
- `services/payroll/contract.service.ts` validates employee ownership, effective-date order, overlapping active contracts, signed-evidence requirements for active contracts, and payroll eligibility.

Assessment:

The foundation is strong for payroll eligibility and contract evidence. A full HRIS needs a broader employment lifecycle around it: draft, review, approval, signature, probation, renewal, amendment, suspension, termination, offboarding, retention, and legal hold. Contract operations also need a consistent request-review-approve-apply pattern rather than allowing all privileged users to directly perform every high-impact action.

Data caution:

Contract overlap appears to be enforced primarily in application logic. For enterprise readiness, PostgreSQL constraints or equivalent transactional protections should prevent concurrent overlapping active ranges at the database layer.

3.3 Compensation And Payroll Input Controls

Stoquify has compensation/rubrique logic, salary-change workflows, maker-checker behavior, evidence hashes, requester/approver separation, and active-contract gating.

Evidence:

- `PayrollEmployeeRubriqueAssignment` begins at `prisma/schema.prisma:1786`.
- `services/payroll/compensation.service.ts` manages salary requests and compensation assignments, including separation between requester and approver/apply actors.
- `actions/payroll/payroll-compensation.actions.ts` applies permission and fresh-auth checks.

Assessment:

This is important, but compensation is broader than payroll input. Full HRIS compensation should include grades, bands, allowances, benefits, deductions, approvals, eligibility, effective dating, employee-facing visibility rules, and audit-ready change history. Country-pack and accounting boundaries must still own statutory meaning and posting mappings.

Data caution:

Compensation assignments should eventually link to explicit contract/employment versions and should have overlap protection. Payroll rubriques should remain calculation definitions; they should not become the sole employee compensation truth.

3.4 Payment Destination Evidence

The current system supports payroll payment-destination change requests with masked/hash destination fields, approval/rejection/application actors, reasons, evidence hashes, and business events.

Evidence:

- `PayrollPaymentDestinationChangeRequest` begins at `prisma/schema.prisma:1877`.

- `services/payroll/payment-evidence.service.ts` handles readiness, masking, hashes, evidence matching, audit, and business events.

Assessment:

This is a strong control surface. A full HRIS should keep raw payment details out of ordinary list payloads and reports. Sensitive values should remain masked, encrypted when recoverable, hash/HMAC-addressed when used as evidence, and subject to fresh authentication for reveal/export.

Security caution:

Deterministic unkeyed SHA-256 is not sufficient for low-entropy values such as phone numbers, account fragments, or national identifiers. Use purpose-specific HMAC-SHA-256 with secret rotation for equality checks, and encryption with managed keys when the value must be recoverable.

3.5 Attendance Snapshots

The `PayrollAttendanceSnapshot` model stores aggregate payroll-period facts: scheduled, worked, overtime, absence, and leave minutes, source hashes, freeze metadata, and correction linkage.

Evidence:

- `PayrollAttendanceSnapshot` begins at `prisma/schema.prisma:1970`.

Assessment:

This is not a full time and leave system. It is a payroll input snapshot. A proper HRIS needs calendars, schedules, holidays, leave policies, leave balances, requests, approvals, overtime workflows, timesheet imports, attendance corrections, and certified period snapshots that payroll consumes.

Data caution:

Attendance snapshots currently store period totals and source payload metadata. Full HRIS requires source-level time/leave facts, not only aggregate payroll inputs.

3.6 Payslip Self-Service

Employee self-service exists primarily through payslip retrieval/export.

Evidence:

- `actions/payroll/payroll-payslip-self-service.actions.ts` resolves own payslip access and uses fresh authentication for export.

Assessment:

This is useful but narrow. Full HRIS self-service should include own profile, employment summary, document requests/status, time and leave, payment-destination change request, and a controlled view of payslips and payroll history.

3.7 Manager Scope

A manager-scope service exists, based on `Location.managerId` and employees assigned to managed locations.

Evidence:

- `services/payroll/org-manager-scope.service.ts` resolves manager-scoped employees.

Assessment:

This should be labeled honestly as managed locations/teams. It is not yet a true reporting-line model. A full HRIS needs effective-dated org units, positions, jobs, assignments, reporting lines, delegations, and manager history.

3.8 Accounting And Assurance Chain

Recent payroll assurance work links certified HRIS input proof to payment/declaration proof and accounting close/data-trust controls. Reports indicate that close/data trust can fail closed when certified HRIS input proof is absent.

Evidence:

- `what-next/payroll/AQSTOQFLOW_HRIS_PAYROLL_PAYMENTS_DECLARATIONS_PROOF_2026-07-14.md`
- `what-next/payroll/AQSTOQFLOW_HRIS_PAYROLL_ACCOUNTING_CLOSE_ASSURANCE_2026-07-14.md`
- `services/accounting/data-trust.service.ts`
- `services/accounting/close-assurance-pack.service.ts`

Assessment:

This is the right architecture direction: HR truth should certify payroll inputs; payroll should generate money/statutory facts; accounting and assurance should validate traceability. The remaining weakness is that some controls are still in current working-tree reports and are not a substitute for production-certified, committed, release-tied evidence.

3.9 Current Security Foundation And Gaps

Observed strengths:

- Payroll actions generally derive tenant and actor context from verified session flows.
- Sensitive payroll services usually include `organizationId` in queries.
- Compensation and payment-destination changes include maker-checker-style workflows.
- Redaction policies cover salary, documents, compliance payloads, provider references, and audit context.
- Payroll mutations commonly write audit records and business events.
- Repository gates already cover many policy, migration, browser, and release-evidence checks.

Observed gaps:

- Fresh authentication appears to be session-recency based rather than proven step-up MFA. A newly stolen session could still satisfy sensitive-operation freshness during the allowed window.
- Tenant isolation is mostly application-query discipline. A future query that omits `organizationId` could bypass the intended tenant boundary unless database row-level security or equivalent controls are added.
- OAuth/session/invite/verification tokens and sensitive identifiers need stronger at-rest protection, including token digests, envelope encryption for recoverable secrets, and keyed hashes for identifier equality.
- Generic audit records are flexible and useful, but not yet uniformly append-only, hash-chained, retained, or exported to immutable storage.

- Segregation of duties is strong in some workflows and incomplete in others, especially termination, contract lifecycle, payment approval/release separation, and broad HR lifecycle changes.
- HR document storage, binary download authorization, retention, legal hold, malware scanning, and verifiable deletion are not yet enterprise-complete.
- Flexible JSON metadata can bypass redaction if sensitive keys are added outside a strict schema.
- Release readiness should fail closed on enterprise-security gates such as real step-up MFA, database tenant isolation, document security, retention/legal hold, audit integrity, key rotation, and cross-tenant negative testing.

4. What HRIS Is Today

Today, Stoquify's HRIS is not a standalone module. It is an HR/payroll source-truth spine expressed mostly through payroll tables, payroll services, payroll actions, and payroll dashboard routes.

Current practical definition:

- Employee identity exists through `PayrollEmployee`.
- Employment contracts exist through `PayrollContract`.
- Compensation and deductions exist through payroll rubrique assignments and salary-change workflows.
- Payment-destination evidence exists through payroll payment evidence services.
- Attendance exists as payroll-period snapshots.
- Employee self-service exists mainly for payslips.
- Manager scope exists through location-managed employees.
- Payroll readiness, payment/declaration proof, and close assurance are increasingly tied to certified HRIS input hashes.

What this means operationally:

- Stoquify can already support controlled payroll workflows with meaningful controls.
- It cannot yet support complete HR operations from hire to retire.
- It should not claim unrestricted enterprise HRIS readiness.
- It should avoid building more HR truth inside payroll-specific services without a boundary plan.

5. What Full Enterprise HRIS Requires

A full Stoquify HRIS should become the system of record for people, employment lifecycle, organization structure, time/leave, documents, HR approvals, and employee/manager self-service. Payroll should consume HRIS-certified snapshots rather than directly owning mutable HR facts.

5.1 Required Domain Boundaries

Boundary	Owns The Truth For	Should Not Own
HRIS / People	Employee identity, employment lifecycle, org assignments, contracts, HR documents, time/leave approval, payment-destination approval, HR approvals	Payroll calculations, statutory formula meaning, ledger posting

Boundary	Owns The Truth For	Should Not Own
Payroll	Periods, readiness verdicts, certified input snapshots, payroll runs, run lines, payslips, corrections, payment batches, payroll declarations	Mutable employee master truth
Country Pack / Compliance	Statutory formula provenance, effective windows, legal mappings, certified fixtures	Employee personal master records
Accounting	Journal posting, close, reconciliation, accounting assurance, close-pack certification	HR profile editing
Auth / RBAC	User identity, session security, tenant membership, permissions, fresh-auth policy	Business truth
Security / Privacy	Redaction, sensitive access audit, secret management, retention, legal hold, export controls	Operational HR decisions

5.2 Required Functional Capabilities

1. Employee master and identity management.
2. Employment lifecycle: onboarding, probation, transfer, promotion, suspension, termination, offboarding, rehire.
3. Organization structure: legal entity, locations, departments, jobs, positions, reporting lines, manager delegations.
4. Contract lifecycle: draft, evidence, review, approval, signature, renewal, amendment, termination.
5. Compensation and benefits: grades, ranges, allowances, benefits, deductions, eligibility, effective dating, approvals.
6. Time and attendance: schedules, calendars, holidays, time entries, anomalies, corrections, approvals.
7. Leave management: policies, balances, accruals, requests, approvals, team calendar, payroll impact.
8. HR documents: evidence register, document types, validity, expiry, retention, legal hold, redacted access.
9. Employee self-service: own profile, docs, leave, attendance, payslips, payment-destination change requests.
10. Manager self-service: scoped roster, leave/attendance approvals, onboarding/offboarding tasks, team readiness.
11. HR analytics and movement history: employee movements, contract movements, comp changes, leave movements, payroll-input certification history.
12. Compliance and audit: SoD, maker-checker, immutable evidence, release attestation, audit export, data retention.

5.3 Capabilities To Defer

The following are valuable, but should not distract from the core HRIS/payroll trust chain:

- Full recruitment/ATS.
- Performance management.
- Learning management.
- Succession planning.
- Advanced workforce planning.
- Benefits marketplace integrations.

They should come after the employee master, lifecycle, time/leave, document evidence, self-service, and payroll certification chain are stable.

5A. Product Strategy: Payroll-Ready People Core

The honest product wedge is not a generic full HR suite. The defensible next product is a Payroll-Ready People Core: authoritative employee, organization, contract, compensation, document, payment-destination, and attendance-input truth that produces a certified payroll snapshot.

This fits Stoquify's existing strength: trusted operational evidence flowing from people data through payroll, cash movement, accounting, and close. A broad generic HRIS would dilute that advantage and delay higher-risk payroll, cash, reconciliation, and accounting production work.

Recommended current label:

Stoquify is an OHADA business operating system with a controlled-pilot payroll kernel and payroll-ready people-data controls.

Do not market Stoquify as a full enterprise HRIS until the HRIS boundary, time/leave source engine, lifecycle workflows, self-service, manager workflows, document retention, live statutory/provider evidence, and release attestation are complete.

MVP Scope

The MVP should be People Core, not full HRIS. It should include:

Capability	MVP Requirement
Employee master	Create/import/update employee identity, employment status, user mapping, duplicate detection, lifecycle timeline
Organization structure	Durable location, department, position, manager, cost-center references, effective dates, scoped manager access
Contract and compensation	Operator forms for contracts, evidence, amendments, approved salary and rubrique changes
Documents	Metadata/evidence center with document type, hash, expiry, retention class, access policy, secure download boundary
Payment destination	Employee request plus maker-checker approval; masked/hash-only operational views
Attendance input	Approved import or integration, freeze, source hash, correction diff; not full scheduling yet
Readiness	Employee, branch, and period queues explaining exactly why payroll is blocked
Payroll handoff	Certified immutable snapshot consumed by the existing payroll engine
Basic self-service	Own profile, payslips, correction request, document request/upload, payment-destination request
Implementation tooling	CSV import, dry-run validation, duplicate report, migration diff, rollback/correction plan

MVP release gates:

- Every included payroll employee maps to one tenant-scoped employee identity.
- Every payroll run consumes a certified snapshot; no UI-supplied readiness claims.

- No draft or unapproved contract, compensation, attendance, or payment destination reaches payroll.
- Managers cannot read outside assigned scope.
- HR/payroll writes have user-facing workflows, not only backend actions.
- A pilot tenant completes two payroll cycles without destructive data repair.

Sequencing Against Cash, Accounting, And Payroll

Cash and accounting should retain priority over broad HRIS expansion because they are closer to Stoquify's current market promise and carry immediate money-movement and close risk.

Recommended order with one delivery squad:

1. Release integrity plus cash/reconciliation/close external certification.
2. Payroll-Ready People Core.
3. Cameroon production payroll and migration pilot.
4. Time, leave, and self-service.
5. Broader HRIS.

With two squads, run cash/accounting final-mile work and the narrow People Core in parallel, then converge at the payroll production pilot. Do not divert either squad into recruitment, performance, or AI before that convergence.

Commercial Packaging

Near term, sell Payroll Pro with People Core included rather than a separate HRIS entitlement. Do not introduce a separately priced **hris** module until People Core is real and durable package/entitlement truth exists.

Recommended commercial evolution:

Offer	Contents	Pricing Metric
People Core	Employee/org/contract/documents/basic self-service	Platform base plus active employees/month
Time & Attendance	Schedules, leave, overtime, approvals, attendance integrations	Active employees plus optional integration/location fee
Payroll Pro	People Core, certified snapshots, runs, payslips, statutory pack	Active payroll employees plus legal entity/country pack
Payroll Assurance	Payroll Pro plus payments, declarations, reconciliation, accounting close evidence	Enterprise bundle plus implementation/integration fee
Talent Suite	Recruitment, performance, learning	Deferred add-on

Commercial cautions:

- Do not charge employees for self-service seats; adoption improves data quality.
- Charge separately for migration, implementation, country-pack certification, and external integrations.
- Live payroll should be an assisted pilot, not a self-serve free trial.
- Downgrade should stop new writes, payroll runs, payment release, and declarations while preserving redacted payslips, registers, and legal evidence read-only.

- Avoid publishing numeric prices until customer employee counts, willingness to pay, implementation cost, and competitor replacement budgets are measured.

What Not To Build Yet

Do not prioritize these before People Core and payroll/cash/accounting certification are stronger:

- Recruitment/ATS.
- Performance reviews.
- Learning management.
- Succession and engagement surveys.
- AI HR copilot or workforce predictions.
- Generic executive HR dashboards.
- Full benefits marketplace.
- Native biometric hardware.
- Deep scheduling integrations before import/adaptor demand is proven.
- Multi-country payroll beyond the first fully certified country pack.
- A big-bang rewrite from `services/payroll` into `services/hris`.
- Production automated payments or declarations without real provider/authority certification.
- A new commercial HRIS module slug before durable entitlement and independent demand exist.

Recommended north star:

Percentage of payroll periods completed from certified HRIS snapshots and closed without high-severity payroll evidence exceptions.

Initial pilot measures:

- 100% of payroll runs from certified snapshots.
- At least 95% of employees ready by payroll cutoff after two cycles.
- Zero cross-tenant or out-of-scope manager reads.
- Declining post-issue payslip correction rate.
- Measurable reduction in payroll preparation hours.
- Successful attachment of Finance/Assurance to payroll customers.

Confidence is high on technical sequencing and moderate on market packaging. Before funding broad HRIS phases, run 8-10 interviews across HR, payroll, finance, and employees, then baseline the current spreadsheet-heavy payroll workflow.

6. Proposal By Capability Area

6.1 HRIS Service Boundary

Business use case:

Create one trusted owner for people truth so payroll, accounting, compliance, and dashboards no longer infer HR facts from route-local UI or payroll-only services.

Users benefiting:

HR operators, payroll operators, accountants, managers, employees, auditors, owners, compliance reviewers.

Truth owner:

New `services/hris/*` boundary. Initially it can wrap the existing `PayrollEmployee`, `PayrollContract`, compensation, attendance, and payment-destination tables. Over time, it should introduce dedicated HRIS models where the payroll naming becomes misleading or insufficient.

UI pattern:

Dedicated People/HRIS workspace, not only payroll screens. Recommended route family:

- `/dashboard/people`
- `/dashboard/people/employees`
- `/dashboard/people/employees/[employeeId]`
- `/dashboard/people/organization`
- `/dashboard/people/time`
- `/dashboard/people/leave`
- `/dashboard/people/documents`
- `/dashboard/people/me`
- `/dashboard/people/team`

Value:

Clear ownership, safer payroll integration, cleaner UX, easier audit, better product positioning.

Risks and tradeoffs:

Creating a new boundary too quickly could duplicate payroll logic. The first phase should use facades/read models over existing tables rather than immediate schema churn.

Implementation path:

1. Add HRIS service facade/read models over existing payroll source tables.
2. Add `hris.*` permissions while preserving existing `payroll.*` permissions.
3. Add HRIS route registry and sidebar entries.
4. Move new people-source mutations behind HRIS services.
5. Keep payroll consuming certified snapshots.

Verification:

- Boundary tests proving HRIS services own people mutations.
- Payroll tests proving payroll consumes HRIS-certified inputs.
- RBAC tests proving HR users and payroll users have different capabilities.
- Regression tests proving existing payroll routes still work.

6.2 Employee Master And Identity

Business use case:

Provide one trusted employee profile for operational HR, payroll readiness, access mapping, manager scope, and audit.

Users benefiting:

HR admins, payroll admins, managers, employees, auditors.

Truth owner:

HRIS employee service. Current storage can remain `PayrollEmployee` during migration.

UI pattern:

Employee directory plus Employee 360 profile. Use compact tables, privacy-safe columns, command brief, status strip, action queue, evidence/proof strip, and detail drawers for sensitive sections.

Default directory columns:

- Employee
- Status
- Job
- Department
- Location
- Readiness
- Evidence status

Sensitive fields such as salary, legal identifiers, bank data, tax identifiers, and raw documents should not be included in list payloads.

Value:

Reduces payroll setup errors, makes HR operations visible, improves audit readiness, and gives managers a reliable team view.

Risks and tradeoffs:

If employee data stays payroll-named forever, future HR features will feel bolted on. If schema migration happens too early, it may destabilize payroll. Use a facade-first approach.

Implementation path:

1. Create HRIS employee read model.
2. Add employee profile page with permission-gated tabs.
3. Resolve employee-user mapping server-side.
4. Add lifecycle timeline assembled from audit/business events.
5. Add redaction-aware drawers for sensitive sections.

Verification:

- Tenant isolation tests.
- User mapping conflict tests.
- Redaction tests for list payloads and rendered DOM.
- Own-record and manager-scope negative tests.

6.3 Employment Lifecycle

Business use case:

Manage onboarding, probation, transfer, promotion, suspension, termination, offboarding, and rehire as traceable HR workflows instead of ad hoc employee record updates.

Users benefiting:

HR operators, managers, payroll operators, auditors, employees.

Truth owner:

HRIS lifecycle service.

UI pattern:

Action queue and lifecycle timeline. Each lifecycle event should show state, risk, action, proof, effective date, actor, evidence, and downstream payroll impact.

Value:

Reduces missed payroll changes, prevents undocumented terminations/transfers, and creates audit history.

Risks and tradeoffs:

Lifecycle workflow can become too heavy for SMB users. Start with a minimum useful state machine: draft, submitted, approved, effective, cancelled, corrected.

Implementation path:

1. Add lifecycle event/request model or metadata-backed workflow if schema must stay narrow.
2. Tie lifecycle actions to employee status, contracts, assignments, document checklist, and payroll readiness.
3. Add onboarding/offboarding task queues.
4. Add correction-only behavior for finalized payroll periods.

Verification:

- State-transition tests.
- Approval and SoD tests.
- Payroll readiness tests for starter/leaver edge cases.
- Audit evidence tests.

6.4 Organization, Positions, And Manager Scope

Business use case:

Represent who works where, who manages whom, and what organizational structure applied at a point in time.

Users benefiting:

HR admins, managers, owners, payroll admins, auditors.

Truth owner:

HRIS organization service.

UI pattern:

Initially: managed locations and team rosters. Later: effective-dated org tree, position list, reporting-line history, and manager workspace.

Value:

Improves access boundaries, approvals, workforce visibility, leave planning, and audit of manager authority.

Risks and tradeoffs:

The current manager scope is location-based, not a reporting-line hierarchy. Calling employees "direct reports" today would overstate the model.

Implementation path:

1. Keep existing location manager scope and label it accurately.
2. Add organization units, jobs, positions, assignments, and reporting lines.
3. Make all manager scope effective-dated.
4. Add delegations and temporary approver coverage.

Verification:

- Manager route-access tests.
- Negative tests for employees outside scope.
- Effective-date tests for historical manager access.
- Privacy tests ensuring managers cannot see salary, bank, or identifiers unless explicitly authorized.

6.5 Contract And Document Evidence

Business use case:

Control employment agreements, signed evidence, amendments, renewals, expiries, and retention obligations.

Users benefiting:

HR admins, legal/compliance, payroll admins, auditors, employees.

Truth owner:

HRIS contract and document-evidence services.

UI pattern:

Contract workbench and document evidence register. Use compact tables, proof badges, expiry/risk status, review queues, and sensitive detail drawers.

Value:

Prevents payroll from running against unsigned, expired, or unapproved employment terms.

Risks and tradeoffs:

Document storage is a privacy and retention risk. Metadata and hashes are not a full document repository. Raw files need separate authorization, short-lived access, encryption, legal hold, and deletion controls.

Implementation path:

1. Wrap existing contract model with HRIS service.
2. Add contract approval workflow for high-impact actions.
3. Add document metadata model: type, category, version, validity, expiry, owner, retention class, evidence hash, storage pointer.
4. Add document access decisions and redaction reasons.

Verification:

- Contract overlap and active-contract tests.
- Evidence-required tests.

- Document access negative tests.
- Retention/legal-hold tests.
- Export redaction tests.

6.6 Compensation, Benefits, And Payroll Inputs

Business use case:

Manage pay changes, allowances, deductions, benefits, and payroll input eligibility with approval evidence and clean handoff to payroll.

Users benefiting:

HR, payroll, finance, managers, employees, auditors.

Truth owner:

HRIS owns employee-level compensation assignments and approvals. Country pack/compliance owns statutory meaning. Accounting owns posting maps. Payroll consumes approved inputs.

UI pattern:

Compensation control surface inside People/HRIS, permission-gated and redacted by default. Payroll should show certified input status, not become the editor for HR compensation truth.

Value:

Reduces wrong payslips, wrong declarations, payroll disputes, and ledger mismatches.

Risks and tradeoffs:

Compensation touches sensitive personal data and statutory obligations. Overexposure in UI is a high privacy risk.

Implementation path:

1. Keep salary-change maker-checker.
2. Add approval workflow coverage for active rubric assignments and high-impact compensation changes.
3. Link assignments to contract/effective period.
4. Generate certified compensation input snapshots for payroll.
5. Keep list payloads salary-free unless specifically authorized.

Verification:

- Maker-checker tests.
- Incompatible-role tests.
- Sensitive field absence tests.
- Payroll input hash tests.
- Country-pack fixture tests.

6.7 Time, Leave, And Attendance

Business use case:

Manage attendance, overtime, absence, leave requests, balances, and payroll-period certification.

Users benefiting:

Employees, managers, HR, payroll, finance, auditors.

Truth owner:

HRIS time/leave/attendance service.

UI pattern:

Separate time and leave operations from the existing payroll readiness screen:

- Attendance period summary.
- Exceptions and anomaly queue.
- Leave balances and requests.
- Manager approval queue.
- Team calendar.
- Certified payroll-input status.

Value:

Prevents payroll from calculating against mutable or unapproved time data.

Risks and tradeoffs:

Time and leave rules vary by country, contract, schedule, and company policy. A rushed implementation may create legal and payroll errors.

Implementation path:

1. Add leave policy, balance, request, approval, and accrual contracts.
2. Add schedule/calendar/holiday support.
3. Add attendance import and correction chains.
4. Freeze approved period snapshots.
5. Have payroll consume only approved, frozen, traceable snapshots.

Verification:

- Leave accrual and balance tests.
- Manager approval tests.
- Retroactive correction tests.
- Payroll readiness fail-closed tests.
- Browser tests for employee and manager workflows.

6.8 Employee Self-Service

Business use case:

Let employees safely access their own HR/payroll facts without exposing other employees or trusting client-provided employee IDs.

Users benefiting:

Employees, HR support, payroll support.

Truth owner:

HRIS self-service resolver. Payroll remains owner of immutable payslip data.

UI pattern:

`/dashboard/people/me` with tabs for profile, employment, documents, leave/time, payment destination requests, and payslips.

Value:

Reduces HR support load, improves employee trust, and creates auditable self-service workflows.

Risks and tradeoffs:

Self-service can leak sensitive data if it trusts client IDs or preloads hidden fields. Own-employee resolution must be server-side.

Implementation path:

1. Resolve own employee from authenticated `userId`.
2. Add read-only own profile and employment summary.
3. Add document request/status.
4. Add leave/time view after time/leave contracts exist.
5. Reuse existing payslip self-service with fresh-auth export.

Verification:

- Own-record negative tests.
- Tenant-switch tests.
- Sensitive export fresh-auth tests.
- Rendered DOM redaction tests.

6.9 Manager Self-Service

Business use case:

Give managers the ability to act on team HR workflows within approved scope.

Users benefiting:

Managers, HR, payroll, employees.

Truth owner:

HRIS manager service with effective-dated scope. Current phase can use location manager scope.

UI pattern:

`/dashboard/people/team` with team roster, availability, leave/attendance approvals, onboarding/offboarding tasks, document task status, and payroll-readiness signals without sensitive compensation or identifier exposure.

Value:

Moves HR work closer to the responsible manager while preserving access boundaries.

Risks and tradeoffs:

Manager scope is easy to overstate. Current scope is location-based. Reporting-line authority requires a real org model.

Implementation path:

1. Expose location-scoped team roster.
2. Add operation-specific manager permissions.

3. Add approval queues for leave/attendance and lifecycle tasks.
4. Later introduce reporting-line assignments and delegations.

Verification:

- Scope tests by location and effective date.
- Negative tests for out-of-scope employees.
- Sensitive field absence tests.
- Accessibility and mobile tests for queue workflows.

6.10 HR Movement And History Tables

Business use case:

Provide audit-grade history of employee movements and HR/payroll source changes.

Users benefiting:

HR, payroll, managers, auditors, accountants, owners.

Truth owner:

HRIS movement read service assembled from immutable audit logs, business events, lifecycle events, contract changes, compensation changes, payment-destination changes, attendance freezes/corrections, payroll snapshots, and payslip events.

UI pattern:

Movement-style analytical table similar in spirit to the existing inventory movements page, but adapted for HR privacy:

- Compact table.
- Event type, employee, effective date, recorded date, actor, source, risk, proof status.
- Filters by event type, employee, department/location, period, status, proof state.
- Detail drawer for evidence and before/after state.
- Redacted values by default.

Value:

Makes HR changes traceable, improves payroll dispute resolution, and supports audits.

Risks and tradeoffs:

Movement tables can accidentally expose sensitive before/after salary, identifier, or bank data. The table must use redacted summaries and lazy-load sensitive detail only after authorization.

Implementation path:

1. Create movement read model.
2. Normalize events across employee, contract, compensation, payment destination, attendance, and payroll snapshots.
3. Add proof badges and redaction reasons.
4. Add export only after retention/export controls are defined.

Verification:

- Event completeness tests.
- Redaction tests.

- Export permission tests.
- Filter and pagination tests.

7. Security, Privacy, And Control Recommendations

7.1 Security Posture

The target should be defense-in-depth, tamper-evident, privacy-preserving, and audit-ready. Avoid absolute claims such as "bulletproof." No HRIS is bulletproof; the practical goal is strong preventive controls, fast detection, provable accountability, and controlled recovery.

7.2 Access Control

Recommendations:

- Add dedicated `hris.*` permissions instead of overloading `payroll.*`.
- Separate HR admin, payroll admin, manager, employee, accountant, auditor, and owner roles.
- Enforce tenant isolation in service code, not only UI.
- Require fresh authentication for sensitive reveal, download, export, approval, and destructive/corrective actions.
- Add an incompatible-role matrix for segregation of duties.
- Require distinct preparer, approver, and releaser for high-risk workflows above configured thresholds.
- Add periodic access recertification and expiring break-glass access.

7.3 Redaction And Sensitive Data Handling

Recommendations:

- Never include salary, legal identifiers, bank details, tax identifiers, social identifiers, or raw document contents in default list payloads.
- Fetch sensitive sections lazily after server authorization.
- Show `Masked` or `Redacted` states with policy and reason.
- Keep public/customer-facing or low-privilege payloads aggregate or masked.
- Add rendered-DOM tests to prove hidden fields are not simply hidden after being sent to the browser.

7.4 Hashing, HMAC, Encryption, And Tokens

Hashing is useful in HRIS for evidence fingerprints, integrity checks, non-reversible comparison of normalized values, and audit proof. It does not encrypt data, does not prove who created a value, does not protect low-entropy identifiers by itself, and does not replace authorization.

Recommended distinctions:

Need	Recommended Control
Password storage	Argon2id preferred; bcrypt or scrypt acceptable when operationally supported
Evidence fingerprint for documents/snapshots	SHA-256 or SHA-512 over canonical bytes, with stored algorithm/version
Low-entropy identifier matching	HMAC-SHA-256 with secret rotation, not plain SHA-256

Need	Recommended Control
Payment destination proof	Masked display plus keyed hash/HMAC; encrypt recoverable destination values with managed keys
Audit integrity	Append-only records, previous-record/content hash chain, correlation IDs, periodic verification
Session/API tokens	High-entropy random tokens or signed tokens with expiry, audience, issuer, rotation, and replay controls
Export/download access	Short-lived signed URLs or signed access grants with audit and expiry

7.5 Audit, Retention, And Legal Hold

Recommendations:

- Make audit and certification records append-only.
- Add tamper-evident hash chains for sensitive audit and certification streams.
- Add retention classes by evidence type and jurisdiction.
- Add legal hold and approved disposition workflows.
- Avoid cascade-deleting evidence records that may be needed for labor, tax, or audit disputes.
- Add purge certificates and backup/restore retention tests.
- Export audit/security events to SIEM or a monitored evidence sink.

7.6 Production Release Blockers

Unrestricted production should remain blocked until:

- Country-pack approvals cover every enabled statutory family and effective window.
- Live payment-provider settlement evidence and authority acceptance evidence pass.
- Retention, legal hold, and tamper-evident archive controls operate successfully.
- Contract, termination, compensation, attendance, payment, declaration, and close workflows meet the approved SoD matrix.
- Historical assignment/run backfill and reconciliation have signed results.
- Authenticated browser, accessibility, mobile, and RBAC-negative tests pass on the release build.
- A clean committed build produces a signed, current final-readiness attestation tied to commit SHA, environment, tenant/country scope, evidence freshness, and functional owner signoff.

8. UI/UX Proposal

8.1 Dedicated People Workspace

Create a dedicated People/HRIS workspace beside payroll.

Recommended route:

- Use `/dashboard/people` for the product-facing HRIS workspace.
- Keep `/dashboard/payroll/*` stable for payroll runs, payment, statutory, payslip, and close controls.

Reason:

Users should not have to understand payroll readiness screens to manage HR operations. Payroll should remain the certified calculation and payment engine, while People/HRIS becomes the workplace for employee truth, lifecycle, organization, time, leave, documents, and approvals.

8.2 Dashboard Design System

All HRIS authenticated screens should follow Stoquify's command-center design system:

- Dark dashboard canvas.
- `.dashboard-landing-theme dark`.
- `.dashboard-landing-content`.
- `--dash-*` tokens.
- Compact, scannable tables.
- Shared primitives: command brief, status strip, KPI/status row, action queue, evidence/proof strip, workbench table, and detail drawer.
- Every screen must answer: What is the state? What is the risk? What is the action? What is the proof?

Avoid:

- Decorative dashboards.
- Route-local palettes.
- Duplicated cards.
- Raw business truth invented in UI components.
- Loading sensitive data into the DOM and hiding it visually.

8.3 Core Screens

People dashboard:

- Workforce state: active employees, starters, leavers, unmapped users, contract readiness.
- Risk: expiring contracts/documents, missing evidence, attendance exceptions, pending approvals.
- Action: onboarding/offboarding tasks, document review, attendance corrections, salary/destination approvals.
- Proof: source freshness, HR evidence coverage, certified payroll-input status.

Employee directory:

- Compact, searchable table.
- Privacy-safe columns.
- Filters by status, location, department, readiness, proof state.
- No raw salary, bank, social, tax, or document values in list payloads.

Employee profile:

- Header with display name, employee number, status, job, department, location, source freshness.
- Tabs for overview, employment, time/leave, documents, compensation, payroll, and history.
- Compensation and identifiers permission-gated and redacted by default.

Organization:

- Phase 1 label: Managed locations and teams.
- Later: org units, positions, reporting lines, delegations, historical manager scope.

Time and leave:

- Attendance summaries, exceptions, corrections, freeze status.
- Leave balances, policies, requests, approvals, team calendar.

Document evidence:

- Metadata-first register.
- Validity, expiry, retention class, owner, proof hash, access decision.
- Raw files behind short-lived authorized access.

Self-service:

- My profile.
- My employment.
- My documents.
- My leave/time.
- My payment-destination request.
- My payslips.

Manager workspace:

- Scoped team roster.
- Availability and exceptions.
- Approval queues.
- Onboarding/offboarding tasks.
- Team readiness without sensitive salary/bank/identifier exposure.

9. Data Architecture Proposal

9.1 Phase 1: Facade Before Schema Split

Do not immediately rename or migrate every payroll table. Introduce an HRIS facade first:

- `services/hris/employee.service.ts`
- `services/hris/lifecycle.service.ts`
- `services/hris/org.service.ts`
- `services/hris/contract.service.ts`
- `services/hris/compensation.service.ts`
- `services/hris/time-leave.service.ts`
- `services/hris/document-evidence.service.ts`
- `services/hris/movement-history.service.ts`

This lets the product create the correct ownership boundary without destabilizing payroll.

During this phase, avoid duplicate truth:

- Keep the existing employee row as the compatibility identity record.
- Make HRIS services the only writer for people-source facts.
- Treat payroll snapshots as immutable evidence copies, not editable HR truth.

- Block direct legacy writes after cutover with database privileges, service checks, or triggers.
- Compare canonical-versus-projection hashes before removing compatibility columns.

9.2 Phase 2: Dedicated HRIS Models

After the facade is stable, add dedicated models where payroll-named tables are insufficient:

- `HrisEmployee` or a mapped/renamed canonical employee identity, without creating a parallel employee master.
- `HrisEmployeeUserLink`
- `HrisEmployment`
- `HrisEmploymentVersion`
- `HrisLegalEmployer`
- `HrisEmployeeLifecycleEvent`
- `HrisOrgUnit`
- `HrisJob`
- `HrisPosition`
- `HrisPositionAssignment`
- `HrisManagerAssignment`
- `HrisCostCenterAllocation`
- `HrisReportingLine`
- `HrisManagerDelegation`
- `HrisContract`
- `HrisContractAmendment`
- `HrisCompensationVersion`
- `HrisCompensationComponentAssignment`
- `HrisBenefitPlan`
- `HrisBenefitEnrollment`
- `HrisPaymentDestinationVersion`
- `HrisDocument`
- `HrisDocumentVersion`
- `HrisDocumentLink`
- `HrisDocumentEvidence`
- `HrisWorkCalendar`
- `HrisSchedule`
- `HrisScheduleAssignment`
- `HrisTimeEntry`
- `HrisLeavePolicy`
- `HrisLeaveBalance`
- `HrisLeaveRequest`
- `HrisLeaveAccount`
- `HrisLeaveEvent`

- `HrisOvertimeRequest`
- `HrisTimeCorrection`
- `HrisAttendanceEntry`
- `HrisAttendanceCorrection`
- `HrisPayrollInputCertification`

Use half-open effective-date ranges for payroll-affecting facts:

```
validFrom <= asOf < validTo
validTo = null means unbounded
```

Use date-only validity for employment, contract, compensation, and assignment facts. Use timestamped event times for approvals, recordings, signatures, imports, and corrections. Corrections should insert new versions and reference the superseded record; they should not rewrite historical content.

Priority database constraints:

- Composite tenant foreign keys such as `(organizationId, employeeId)`.
- Database checks for date ordering, nonnegative amounts/minutes, status/evidence consistency, and balance arithmetic.
- Exclusion or equivalent constraints preventing overlapping active ranges for employment, contracts, position/manager assignments, base compensation, component assignments, schedule assignments, and payment destinations.
- Partial unique indexes for open salary-change requests, open payment-destination requests, current attendance certifications, and active payroll runs.
- FK-leading indexes for child relationships.
- Mandatory idempotency keys after historical nulls are backfilled.

9.3 Phase 3: Payroll Consumption Contract

Payroll should consume only certified HRIS snapshots:

- Employee identity snapshot.
- Active contract snapshot.
- Compensation/rubrique snapshot.
- Payment destination proof hash.
- Attendance/time/leave snapshot.
- Certification hash.
- Source freshness and blockers.
- Corrected-from linkage where applicable.

Payroll should fail closed if required HRIS proof is absent, stale, mismatched, or outside the payroll period.

Migration and backfill risks:

- Reconcile the target database's Prisma migration history before adding HRIS migrations.
- Mark ambiguous legacy rows as `LEGACY_UNVERIFIED`; do not invent historical HR facts from incomplete free-text or JSON records.
- Convert inclusive end-date conventions carefully if moving to half-open date ranges.

- Audit dirty data before cutover: orphan/cross-tenant user and location links, overlapping contracts or assignments, multiple frozen attendance rows, duplicate active runs, active compensation without evidence, and payment destinations without applied approval.
 - Never rewrite posted run lines, payslips, declarations, released payment evidence, or close evidence; add append-only source links or correction records.
 - Use expand/contract migrations, concurrent indexes, `NOT VALID` constraints followed by validation, tenant-batched dry runs, reconciliation hashes, and explicit rollback/correction scripts.
-

10. Roadmap

Phase 0: Boundary And Status Register

Goal:

Make ownership explicit before more HR features are added.

Deliverables:

- Refresh HRIS/payroll status register.
- Add HRIS source-truth map.
- Create `services/hris` facade.
- Add `hris.*` permissions.
- Add route registry entries for People workspace.
- Document payroll consumption contract.

Release gate:

Payroll still works; new people-source mutations are routed through HRIS facade.

Phase 1: Employee Identity And Lifecycle

Goal:

Turn employee identity into an HRIS-owned profile and lifecycle workflow.

Deliverables:

- Employee directory.
- Employee profile.
- Lifecycle event/workflow model.
- Onboarding/offboarding task queue.
- Employee movement history.

Release gate:

Tenant/RBAC/redaction tests pass; payroll readiness still consumes certified HRIS facts.

Phase 2: Organization And Manager Scope

Goal:

Move from location-only manager scope toward effective-dated organization authority.

Deliverables:

- Managed locations/team workspace.
- Org units/jobs/positions.
- Reporting-line assignments.
- Delegation model.
- Manager approval scopes.

Release gate:

Manager negative access tests pass; historical scope is traceable.

Phase 3: Contracts, Documents, Compensation

Goal:

Make contract, document, and compensation controls HRIS-owned and audit-grade.

Deliverables:

- Contract lifecycle approvals.
- Document evidence register.
- Retention/legal hold model.
- Compensation assignment controls.
- Benefit/deduction eligibility.
- Approval completeness and SoD matrix.

Release gate:

Sensitive data redaction, document access, SoD, and evidence tests pass.

Phase 4: Time, Leave, Attendance

Goal:

Create operational time/leave workflows and certified payroll input snapshots.

Deliverables:

- Leave policy/balance/request/approval.
- Attendance import/entry/correction.
- Schedule/calendar/holiday support.
- Certified period snapshot.
- Payroll readiness fail-closed behavior.

Release gate:

Payroll cannot run from mutable or uncertified time/leave data.

Phase 5: Employee And Manager Self-Service

Goal:

Open safe HR workflows to employees and managers.

Deliverables:

- `/dashboard/people/me`

- `/dashboard/people/team`
- Own profile, docs, leave/time, payslips.
- Manager queues for scoped tasks.
- Browser/accessibility/mobile proof.

Release gate:

Own-record and manager-scope negative tests pass; rendered DOM redaction tests pass.

Phase 6: Production Certification

Goal:

Convert controlled pilot readiness into broader production readiness.

Deliverables:

- Qualified country-pack signoff.
- Live provider/authority certification.
- Historical backfill and reconciliation.
- Immutable audit/evidence archive.
- Release attestation tied to commit SHA and environment.
- Accountant/compliance/security/ops signoff.

Release gate:

Unrestricted production remains no-go until all blockers are closed with current evidence.

Phase 7: Extended HRIS

Goal:

Add enterprise breadth after the source-truth spine is stable.

Candidate modules:

- Recruiting/ATS.
- Benefits administration.
- Performance management.
- Learning/training.
- Workforce planning.
- Advanced HR analytics.

Release gate:

Each module must use the same HRIS boundary, RBAC, privacy, evidence, and release governance.

11. Testing And Verification Plan

Minimum release gates for any HRIS production slice:

Gate	Required Verification
Prisma/schema	<code>npm run prisma:validate</code> , migration diff review, backfill dry run

Gate	Required Verification
Type safety	<code>npm run typecheck</code>
Lint/static checks	<code>npm run lint</code> and repository policy gates
HRIS services	Unit tests for tenant isolation, state transitions, redaction, and evidence hashes
Payroll integration	Tests proving payroll consumes certified HRIS snapshots and fails closed without proof
RBAC	Negative tests for HR, payroll, manager, employee, accountant, auditor, owner
Privacy	Tests proving sensitive fields are absent from list responses and rendered DOM
Audit	Tests for append-only behavior, correlation IDs, event completeness, and denied-action logging
SoD	Tests for incompatible roles and requester/approver/releaser separation
UI	Authenticated Playwright smoke at desktop/tablet/mobile widths
Accessibility	Keyboard navigation, focus states, labels, no color-only status, Axe checks
Release	Machine-readable final readiness attestation tied to commit SHA, environment, country/tenant scope, and evidence freshness

12. Honest Production Readiness Statement

Current status:

Stoquify is strong enough to continue controlled pilot work for implemented, evidence-gated payroll workflows. It has more serious controls than a basic HR/payroll CRUD module.

Not ready:

It is not yet ready to be called a full enterprise HRIS or unrestricted production HRIS/payroll system.

Why:

- HRIS truth is still embedded mainly in payroll services/tables.
- There is no standalone `services/hris` or `actions/hris` boundary observed.
- No dedicated People/HRIS route family was observed.
- Time and leave are payroll input snapshots, not a full HRIS time/leave module.
- Self-service is mostly payslip-centered.
- Manager scope is location-based, not a true reporting-line model.
- Retention/legal hold and archive controls need hardening.
- SoD needs a platform-wide incompatible-role matrix.
- Statutory country-pack breadth and live authority/provider certifications remain blockers.
- Browser/accessibility/mobile release evidence must be current and release-tied.
- Several roadmap/report artifacts and current controls may be working-tree evidence rather than deployed proof.

Recommended decision:

Continue, but continue honestly. Build the HRIS/People boundary next, preserve payroll's evidence-gated kernel, and keep unrestricted production blocked until the release gates are satisfied.

13. Immediate Next Actions

1. Create the `services/hris` facade and HRIS source-truth map.
2. Add `hris.*` permissions and separate HRIS roles from payroll roles.
3. Add `/dashboard/people` route shell using existing dashboard primitives.
4. Build employee directory, employee profile, and lifecycle timeline over current data.
5. Add HR movement-history read model with strict redaction.
6. Add manager workspace using current location scope, labeled accurately.
7. Design dedicated document evidence and retention/legal-hold model.
8. Design time/leave contracts before building leave request UI.
9. Tie every payroll run to certified HRIS input proof and fail closed on missing/stale proof.
10. Create release attestation that blocks on dirty/uncommitted artifacts, stale evidence, missing live proof, and absent owner signoffs.

14. Final Recommendation

The product direction is sound: Stoquify should become a full enterprise HRIS plus payroll assurance system. The current codebase already has enough controlled payroll and HR source-data foundation to make that realistic.

The main risk is overclaiming readiness before the HRIS boundary, lifecycle breadth, privacy controls, retention, live statutory/provider evidence, self-service, and release attestation are complete.

The professional path is:

- Do not rewrite payroll.
- Do not keep expanding HR truth inside payroll-only screens.
- Build People/HRIS as the owner of employee truth.
- Let payroll consume certified HRIS snapshots.
- Keep accounting and assurance tied to proof.
- Release in narrow, evidenced slices.

This approach gives Stoquify a credible route to a modern, enterprise-grade HRIS without weakening the controls that already make the payroll foundation valuable.